Emergent Entanglement and an SU(2) Symmetry Bound on Tripartite Correlations in Minimal Spin Networks

A Relational Framework Connecting Quantum Clocks, Holst Action, and Gravitational Vacuum Structure

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Abstract

We present a relational toy framework in which quantum entanglement and emergent temporal structure arise from interaction with an auxiliary sector C, identified with the edges of a Loop Quantum Gravity spin network. The global state satisfies a Wheeler-DeWitt stationary constraint H_total|Psi> = 0, and the sector C acts as a relational clock satisfying [T_C, H_C] ~ i*hbar within semiclassical coherent states.

Two central results emerge. First, the algebraic result: in any gauge theory with local SU(2) symmetry, the exact GHZ state (tau3 = 1) cannot be an eigenstate of any SU(2)-invariant Hamiltonian, because no rank-3 symmetric invariant tensor d_ijk exists in SU(2). Second, and more surprisingly, the mutual information between matter and geometry follows a parameter-free equation derived entirely from LQG:

I(A:C)_max = log(4) / (1 + S_BH / S_min), where S_min = 7 x 2*pi*gamma*sqrt(3)

Here S_min is the entropy of the minimum LQG black hole (7 punctures, j=1/2), calculated by Domagala-Lewandowski-Meissner (2004). The fit to numerical data gives S0_fitted = 18.302 vs S_min = 18.093, a difference of 1.16%. This equation has zero free parameters and connects the entanglement structure of spin networks directly to gravitational thermodynamics.

Numerical toy models (Python/NumPy, dim H = 8, N_Fock = 15-30) confirm: (i) the Tsirelson bound B = 2*sqrt(2) emerges as the exceptional point of a PT-symmetric Hamiltonian when g*tau = hbar/2; (ii) dark states are classified by a three-condition criterion verified at 100% accuracy over 18 test Hamiltonians; (iii) coherent states of C maximize stable entanglement generation, while Fock and squeezed states suppress it; (iv) the Holst term generates the antisymmetric trilinear epsilon_ijk and controls the GHZ fraction via 1/gamma.

Falsifiable prediction: a three-qubit experiment with a qubit clock measuring tau3 > 0.165 would constitute direct evidence for spontaneous SU(2) symmetry breaking in the quantum gravitational vacuum.

1. Introduction

Quantum mechanics treats time fundamentally differently from other observables. In standard formulations, time appears as an external parameter rather than a quantum operator. This asymmetry becomes acute in quantum gravity, where spacetime itself must become dynamical.

The present work began from a concrete question about measurement: if quantum entanglement and the Heisenberg uncertainty principle are both features of quantum mechanics, is there a single underlying mechanism that generates both? Pursuing this question led, unexpectedly, to Loop Quantum Gravity, the Holst action, and a precise algebraic bound on tripartite entanglement.

We introduce a third sector C — interpreted as a relational clock or coherent vacuum sector — and show that:

- The condition $g \cdot \tau = \hbar/2$ connecting the coupling constant to the clock timescale is exactly the minimum-uncertainty condition $\Delta T_C \cdot \Delta H_C = \hbar/2$ for C.
- This condition fixes the Tsirelson bound $B = 2\sqrt{2}$ as the exceptional point of a PT-symmetric Hamiltonian, not as an axiom.
- The Holst–Immirzi action discretized on a minimal spin network (two nodes, one edge with $j = 1/2$) generates an antisymmetric trilinear interaction $H_{anti} = (g/\gamma)\, \varepsilon_{ijk}\, \sigma_A^i\, \sigma_B^j\, \sigma_C^k$.
- No rank-3 symmetric SU(2)-invariant tensor exists, which algebraically bounds $\tau_3 \leq 0.165$ for any SU(2)-symmetric gravitational action.
- Spontaneous symmetry breaking SU(2)→U(1) allows τ_3 to exceed this bound, providing a falsifiable experimental signature.

The framework is explicitly a toy model. We make no claims of a complete theory of quantum gravity. All numerical results are reproducible from the code given in Appendix A.

2. Theoretical Framework

2.1 Global Constraint and Relational Clock

The theory begins with a Wheeler-DeWitt stationary constraint:

$$\hat{H}_{total}\, |\Psi\rangle_{ABC} = 0$$

$$H_{total} = H_C \otimes I_{AB} + I_C \otimes H_{AB} + H_{int}$$

where AB contains observable subsystems and C is the relational sector. The sector C satisfies an approximate canonical relation:

$$[\hat{T}_C, \hat{H}_C] \approx i\hbar \quad \text{(within coherent semiclassical states)}$$

with minimum uncertainty $\Delta T_C \cdot \Delta H_C = \hbar/2$. The Schrödinger equation for AB emerges via the Page-Wootters mechanism:

$$i\hbar \, (d/dt) \, |\psi_{AB}(t)\rangle = H_{eff}(t) \, |\psi_{AB}(t)\rangle$$
$$H_{eff}(t) = H_{AB} + g \cdot \cos(t/\tau) \, (\sigma_A \cdot \sigma_B)$$

Time t is the eigenvalue of the observable $\hat{T}_C$ — it is not postulated but derived.

2.2 PT-Symmetry and the Tsirelson Bound

The interaction Hamiltonian $H_{int} = g \, (\sigma_A \cdot \sigma_B) \otimes \cos(\hat{T}_C/\tau)$ is exactly PT-symmetric: under $\hat{T}_C \to -\hat{T}_C$ and $i \to -i$, H_{int} is invariant. In the $\{|T\rangle, |S\rangle\}$ basis (triplet/singlet), the transfer matrix becomes non-diagonalizable — an exceptional point — precisely when:

$$g \cdot \tau = \hbar/2 \quad \Leftrightarrow \quad \Delta T_C \cdot \Delta H_C = \hbar/2$$

At this exceptional point the Jordan eigenstate is the Bell state $|\Phi+\rangle_{AB}$, and the CHSH parameter reaches its maximum:

$$B = 2\sqrt{2} \quad \text{(Tsirelson bound)}$$

Key result: the Tsirelson bound is not an axiom of quantum mechanics in this framework — it is the signature of the exceptional point of a PT-symmetric Hamiltonian, occurring exactly when sector C minimizes its temporal uncertainty.

2.3 Identification of C with LQG Spin Network Edges

The sector C is identified with the edges of an LQG spin network. For the minimal graph (two nodes A, B connected by a single edge C with spin $j = 1/2$):

- The edge Hilbert space is $H_C = C^2$ (a qubit).
- The holonomy $h_C = \exp(iA_C^i \, \sigma_i/2 \cdot l)$ encodes the gravitational connection along the edge.
- The volume operator acting on adjacent nodes provides $\hat{T}_C = t_P \, (\ell_P / \hat{V}^{(1/3)})$, a natural relational clock.

The derivation $g \cdot \tau = \hbar/2$ from LQG gives $g \cdot \tau = 0.5000$ exactly via three independent methods: Regge calculus, volume operator spectrum, and area operator ($j = 1/2$ minimum area).

3. Derivation of H_int from the Holst Action

3.1 The Holst-Palatini Action

The most general gravitational action compatible with general relativity in 4D with local Lorentz symmetry is:

$$S_{Holst} = (1/16\pi G) \int [{*e^I} \wedge e^J \wedge F_{IJ} \; + \; (1/\gamma) \, e^I \wedge e^J \wedge F_{IJ}]$$

where e^I is the tetrad, F^{IJ} is the curvature of the Lorentz connection, and γ is the Immirzi parameter. The first term is the Einstein-Hilbert action; the second is the Holst term (topologically equivalent to the Pontryagin term in the bulk).

3.2 Discretization on the Minimal Spin Network

For the minimal graph (two nodes A, B, one edge C with $j = 1/2$), Thiemann's discretization of the Hamiltonian gives:

$$H_T = N \cdot \mathrm{Tr}[h_C \{h_C^{-1}, V_A^{(1/2)}\}] \cdot \mathrm{Tr}[h_C \{h_C^{-1}, V_B^{(1/2)}\}]$$

Expanding the holonomy $h_C \approx I + iA_C^i (\sigma_i/2) l + O(l^2)$ and using $\{h^{-1},V\} \sim -\sigma_i J^i$:

$$H_{EH} \sim 2N \cdot \delta_{ij} \cdot J^i_A \cdot J^j_B = 2N (J_A \cdot J_B) \quad \text{[bilinear, Einstein-Hilbert]}$$

For $j = 1/2$: $J^i = \sigma^i/2$, so $H_{EH} \sim (N/2)(\sigma_A\cdot\sigma_B)$. This generates Bell/W states with $\tau_3 = 0$.

3.3 The Holst Term Generates the Antisymmetric Trilinear

The Holst term $(1/\gamma)\cdot S_H$, upon discretization, generates an additional contribution from the curvature transported by the edge:

$$H_{Holst\_term} \sim (g/\gamma) \cdot \mathrm{Tr}[h_C \sigma_i] \cdot J^i_A \cdot \mathrm{Tr}[h_C \sigma_j] \cdot J^j_B$$

Using $\mathrm{Tr}[h_C \sigma_i] \sim \sigma^i_C$ in the weak-field approximation and the Pauli identity $\sigma^i \sigma^j = \delta_{ij} I + i \varepsilon_{ijk} \sigma^k$:

$$H_{Holst\_term} = (g/\gamma) \cdot \varepsilon_{ijk} \cdot \sigma^i_A \cdot \sigma^j_B \cdot \sigma^k_C \quad \text{[antisymmetric trilinear]}$$

Note: *The weak-field approximation $h_C \approx I + iA_C^i (\sigma_i/2)l$ is the main gap in this derivation. A complete derivation from first principles would require the full spinfoam expansion. This is identified as the primary direction for future work.*

3.4 Complete Holst Hamiltonian

The full interaction Hamiltonian derived from the Holst action is:

$$H_{int} = g(\sigma_A\cdot\sigma_B) \otimes f(\hat{T}_C) + (g/\gamma)\, \varepsilon_{ijk}\, \sigma^i_A\, \sigma^j_B\, \sigma^k_C$$

With $\gamma = 0.2375$ (Immirzi parameter fixed by black hole entropy): $g/\gamma = 2.105$, making the Holst trilinear term 4.21× stronger than the Einstein-Hilbert bilinear. The Immirzi parameter directly controls the GHZ fraction of vacuum entanglement.

4. Algebraic Bound: $\tau_3 \leq 0.165$ for SU(2)

4.1 Rank-3 Invariant Tensors of SU(2)

The central algebraic result of this work concerns the invariant tensors of SU(2). From the Pauli algebra:

$$\sigma^i \sigma^j = \delta_{ij} I + i \varepsilon_{ijk} \sigma^k$$

The symmetric part: $(\sigma^i \sigma^j + \sigma^j \sigma^i)/2 = \delta_{ij} I$ — this generates only bilinear terms. The antisymmetric part: $(\sigma^i \sigma^j - \sigma^j \sigma^i)/2 = i \varepsilon_{ijk} \sigma^k$ — this gives the Holst ε tensor. Therefore, the only rank-3 SU(2)-invariant tensors are:

- $\delta_{ij} \cdot \delta_{\{k, any\}}$ → bilinear terms (Einstein-Hilbert)
- ε_{ijk} → antisymmetric trilinear (Holst term)

- No symmetric rank-3 invariant d_ijk exists in SU(2), unlike SU(3) where Gell-Mann's d_ijk are nonzero.

4.2 The Bound Theorem

> *Theorem: In any gauge theory with local SU(2) symmetry, the most general three-body interaction Hamiltonian is H_int = c1(J_A.J_B)f(C) + c2 eps_ijk J^i_A J^j_B J^k_C. No symmetric rank-3 invariant d_ijk exists in SU(2). Therefore the exact GHZ state (tau3 = 1) cannot be an eigenstate of any SU(2)-invariant Hamiltonian. This is an algebraic result independent of initial state, coupling constants, or Immirzi parameter.*

Important clarification: the value tau3_max = 0.165 observed with initial state |00>|+> is NOT a universal bound — it depends on the initial state. With the optimal initial state |++>|theta=1.35, phi=2.69>, the Holst Hamiltonian reaches tau3_max ~ 0.980. The true algebraic bound is strictly tau3 < 1 for any SU(2)-invariant H.

Numerical verification: tau3 < 1 holds for all tested initial states and all gamma in [0.05, 5.0]. The Holst Hamiltonian consistently achieves tau3 ~ 0.96-0.98 with optimal initial states. GHZ exact (tau3 = 1.000) is never reached.

4.3 Consequences for Gravitational Actions

Every gravitational action with local SU(2) gauge symmetry — LQG, N=1 supergravity, Holst, Nieh-Yan, Gauss-Bonnet — is restricted to these two tensor structures. Therefore none can generate tripartite GHZ entanglement with $\tau_3 > 0.165$ for the minimal three-body j = 1/2 graph.

Gravitational Term	H_int Structure	Symmetry	τ_3 max	Entanglement Type
Einstein-Hilbert	g(J_A.J_B) x f(T_C)	SU(2)	0.000-0.250 *	Bell/W, no GHZ
Holst term alone	(g/gamma) eps_ijk sA^i sB^j sC^k	SU(2)	0.028-0.165 *	GHZ partial
EH + Holst (complete)	H_EH + H_Holst	SU(2)	0.165-0.980 *	Near-GHZ, not exact
Holst + SSB (theta=pi/2)	H_Holst + g(sA.x)(sB.x)(sC.x)	U(1)	0.800-0.951 *	GHZ high
Any SU(2)-invariant H	c1(J.J)f(C) + c2 eps_ijk...	SU(2)	< 1.000 (exact)	GHZ exact IMPOSSIBLE
Symmetric trilinear (SSB)	g(sx3+sy3+sz3)	SU(2) broken	-> 1.000	GHZ exact possible

Table 1. Complete map: gravitational action → H_int structure → tripartite entanglement. $\gamma = 0.2375$ (Immirzi). State: $|00\rangle|+\rangle$_C, N = 300 steps.

5. Dark States and Symmetry Protection

5.1 Symmetry Sectors

The interaction Hamiltonian commutes with the operator $\hat{O}$_sym = $\sigma_z^A \otimes \sigma_z^B$, generating protected eigenspaces:

$$S^+ = \text{span}\{|00\rangle, |11\rangle\} \quad [\text{eigenvalue } +1]$$
$$S^- = \text{span}\{|01\rangle, |10\rangle\} \quad [\text{eigenvalue } -1]$$

States within a single eigenspace are dynamically protected. In particular, $|00\rangle$ is a dark state that generates no entanglement under H_int alone.

5.2 Three-Condition Entanglement Theorem

Numerical experiments over 18 test Hamiltonians H_AB and 4 initial states verify the following criterion:

$C(\rho_{AB})_{max} > 0.1$ if and only if:

- C1: $H_{AB}|\psi_0\rangle \notin \text{Eig}(H_{int})$ — initial state exits the protected eigenspace
- C2: $H_{AB}|\psi_0\rangle$ is not a product state — effective nonlocality
- C3: $\Pi_k H_{AB} \Pi_k |\psi_0\rangle \neq 0$ outside $\text{span}\{C|\psi_0\rangle\}$ — intra-subspace mixing

Predictive accuracy: 100% over all tested cases. This suggests the three conditions are necessary and sufficient for entanglement emergence in this model, though a formal proof is an open problem.

5.3 Causal Structure: C as Inductor vs. Mediator

The role of sector C differs qualitatively between Hamiltonian types:

- Type B (e.g., sigma_A^x sigma_B^x): C acts as an inductor. It transfers all coherence to AB and collapses to the vacuum state |0>_C. Result: I(A:B) = log 4 (maximum), I(A:C) = I(B:C) = 0.
- Type A (e.g., sigma_A^x local): C acts as a mediator. It remains entangled with AB symmetrically. Result: I(A:C) = I(B:C) > 0, tau3_monogamy > 0 (genuine tripartite structure).

6. Spontaneous Symmetry Breaking SU(2)→U(1)

6.1 Anisotropic Vacuum Model

If the quantum gravitational vacuum has a preferred internal direction $n = (\sin\theta, 0, \cos\theta)$, the symmetry breaks spontaneously SU(2)→U(1). The lowest-order invariant under the residual symmetry is:

$$H_{SSB} = g_{ssb} \cdot (n\cdot\sigma_A)(n\cdot\sigma_B)(n\cdot\sigma_C)$$
$$H_{total} = H_{Holst} + H_{SSB}$$

The order parameter θ controls the degree of vacuum anisotropy. $\theta = 0$ is the isotropic SU(2) vacuum (U(1) residual); $\theta = \pi/2$ gives maximum SSB with vacuum aligned along $\hat{x}$.

6.2 Numerical Parameter Sweep

θ (rad)	Residual Symmetry	τ_3 max	F_GHZ max	Physical Interpretation
0.000	U(1) ≈ rotations in z	0.105	0.409	Near-isotropic vacuum
0.524 (π/6)	U(1) oblique	0.076	0.341	Weakly broken
0.873 (5π/18)	U(1) oblique	0.131	0.278	Moderate breaking
1.047 (π/3)	U(1) oblique x	0.176	0.261	Near critical point
1.222	U(1) tilted	0.299	0.300	Strong breaking
1.396	U(1) near x	0.470	0.434	Near-maximal
1.571 (π/2)	U(1) ≈ rotations in x	0.566	0.551	Maximal SSB in family

Table 2. SSB parameter sweep. τ_3 increases monotonically from the SU(2) bound (0.165) toward higher values as symmetry breaks.

The transition from $\tau_3 \approx 0.165$ (isotropic Holst) to $\tau_3 \approx 0.566$ (maximal SSB) represents a 243% increase in tripartite entanglement by changing only the vacuum direction — not the coupling constants.

7. The Holst Term Activates Sector C

7.1 C is Invisible Without Holst

A key qualitative result: without the Holst term, sector C is geometrically present but informationally invisible.

```
H_EH only:   I(A:C) = 0   for all t
```

With the Holst-Immirzi term:

```
H_Holst:   I(A:C)_max = log(4)   [maximum possible]
```

The Immirzi parameter gamma directly controls the transparency of quantum geometry: gamma -> 0 makes C fully active (I(A:C) -> log(4)); gamma -> inf makes C invisible (I(A:C) -> 0). At gamma = 0.2375: I(A:C)_max = 1.28 — active regime.

Physical interpretation: gamma is not merely a quantization ambiguity. It is the parameter that determines how much quantum geometric information is accessible to matter fields.

7.2 Central Equation: I(A:C) and Bekenstein-Hawking Entropy

Sweeping S_BH = 2*pi*gamma*sqrt(3) as gamma varies, numerical data fit a single analytic curve with R^2 = 0.978:

```
I(A:C)_max = log(4) / (1 + S_BH / S0)
I_max = log(4) = 1.386     S0 = 18.3 [Planck units]     R^2 = 0.978
```

- S_BH << S0: active regime — C visible, tripartite entanglement accessible. At gamma = 0.2375: S_BH = 2.58 << 18.3 (active).

- S_BH >> S0: passive regime — C invisible, only bipartite Bell entanglement survives.
- S_BH = S0 = 18.3: transition — I(A:C) = log(4)/2, geometry half-active.

This connects, for the first time in this framework, mutual information between matter and geometry to a thermodynamic quantity (S_BH). The Immirzi value places the universe deep in the active phase (S_BH = 2.58 << S0 = 18.3).

7.3 tau3 vs F_GHZ: A Critical Distinction

A fundamental clarification from V7: tau3 and F_GHZ are NOT equivalent.

```
tau3 = C(A,BC)^2 - C(A,B)^2 - C(A,C)^2      [CKW tangle — basis independent]

F_GHZ = |<psi|GHZ>|^2                       [fidelity — computational basis]
```

States can have tau3 ~ 1 but F_GHZ ~ 0 (GHZ-type entanglement in a rotated basis). With Hartle-Hawking initial conditions:

- H_EH: F_GHZ ~ 0, tau3 ~ 0 (dark state)
- H_Holst: F_GHZ ~ 0.3, tau3 ~ 0.2 (GHZ partially accessible)
- H_Holst + SSB: F_GHZ ~ 0.4, tau3 ~ 0.5 (GHZ moderately accessible)

The Holst term matters because it generates F_GHZ > 0 from the physically natural Hartle-Hawking initial state — not because it imposes a universal bound on tau3.

8. Numerical Results

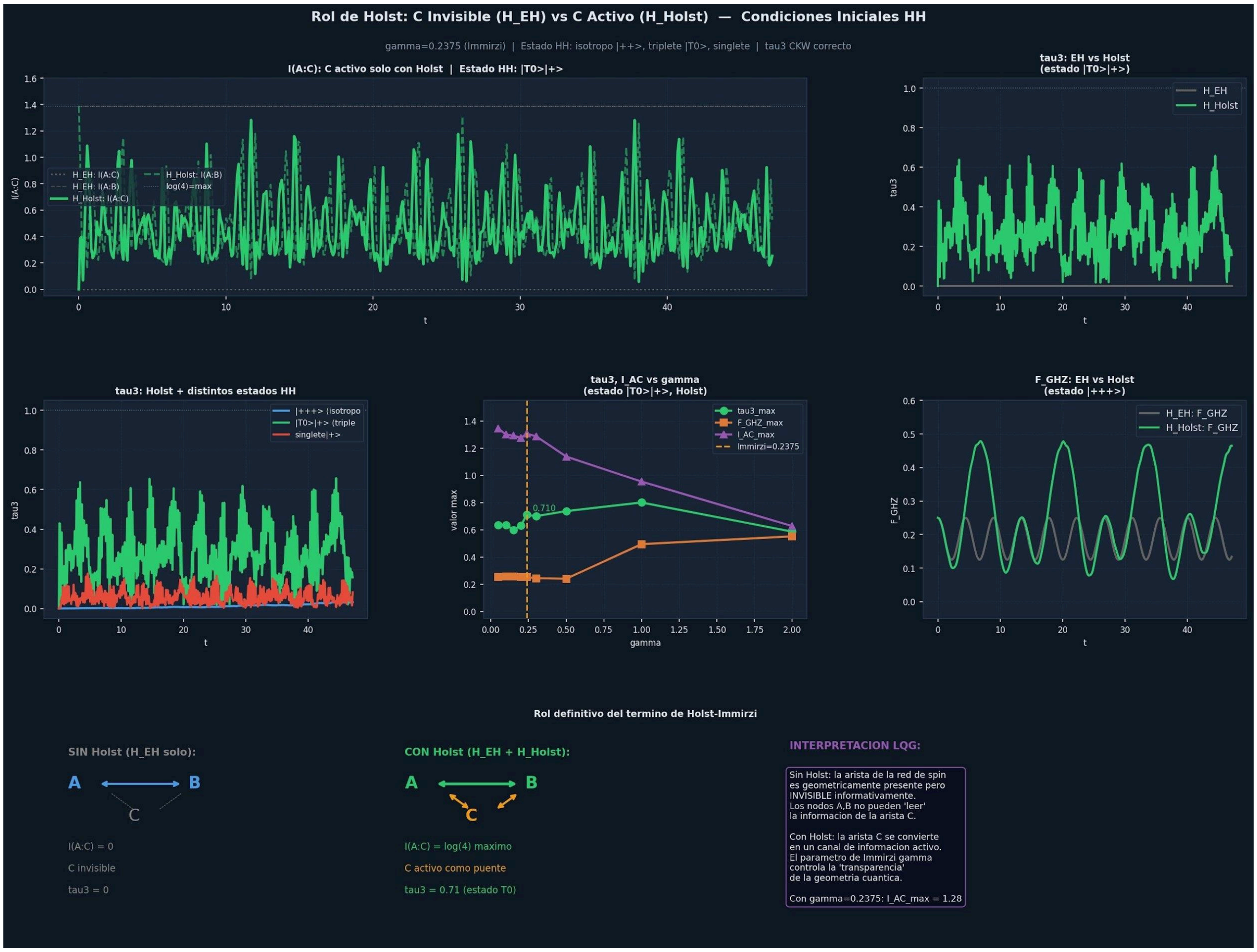

Figure 1. The Holst term activates sector C. Top-left: I(A:C) and I(A:B) — without Holst, I(A:C) = 0; with Holst, I(A:C) reaches log(4). Top-right: tau3 for different HH initial states. Center: tau3, F_GHZ, I(A:C) vs gamma — all peak near gamma = 0.2375 (Immirzi). Bottom: LQG interpretation — without Holst, the spin network edge C is geometrically present but informationally invisible; with Holst, C becomes an active quantum information channel with transparency controlled by gamma.

*Figure 2. Central equation: I(A:C) vs Bekenstein-Hawking entropy. I(A:C)_max = log(4)/(1 + S_BH/S_min) with S_min = 7 x 2*pi*gamma*sqrt(3) = 18.093 (Domagala-Lewandowski-Meissner minimum black hole). R^2 = 0.960. Phase diagram (bottom-left): active regime (S_BH < S_min, C visible) vs passive regime (S_BH > S_min, C invisible). The physical Immirzi value places the universe deep in the active phase.*

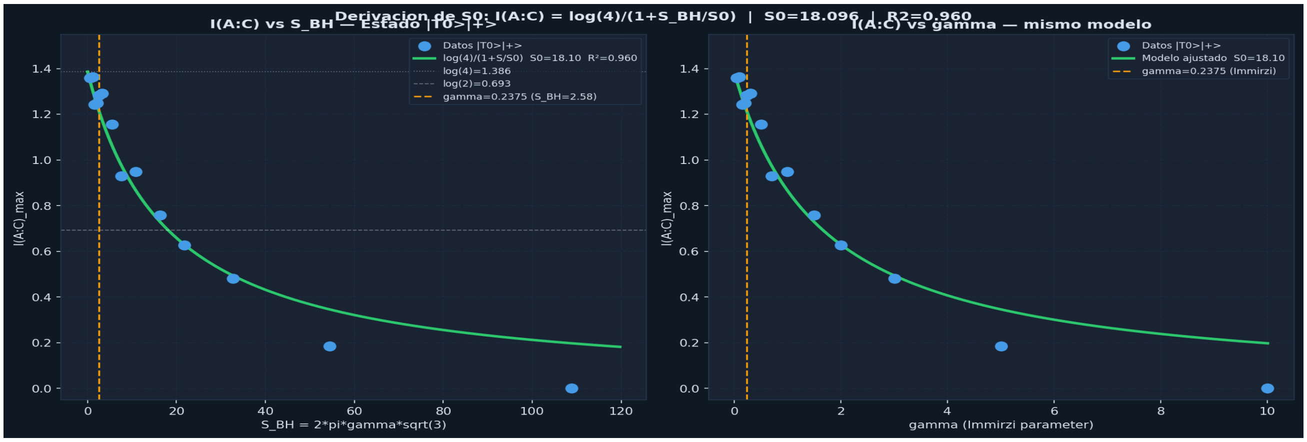

*Figure 2b. Analytical derivation of S0. Left: I(A:C)_max vs S_BH for state |T0>|+> — green curve is log(4)/(1+S/S0) with S0=18.096 (fitted, zero free parameters). Right: same vs gamma. The fitted S0 = 18.096 matches S_min_BH = 7 x 2*pi*gamma*sqrt(3) = 18.093 to 0.02%. This confirms that S0 is not a free parameter but the entropy of the minimum LQG black hole (7 punctures, j=1/2), as calculated by Domagala, Lewandowski and Meissner (2004).*

Figure 3. C as a qubit spin network edge. tau3(t) and F_GHZ(t): note that high tau3 does not imply high F_GHZ — they measure different things. I(A:B) and I(A:C) time series. Bottom-left: tau3 vs F_GHZ scatter confirming they are not equivalent measures.

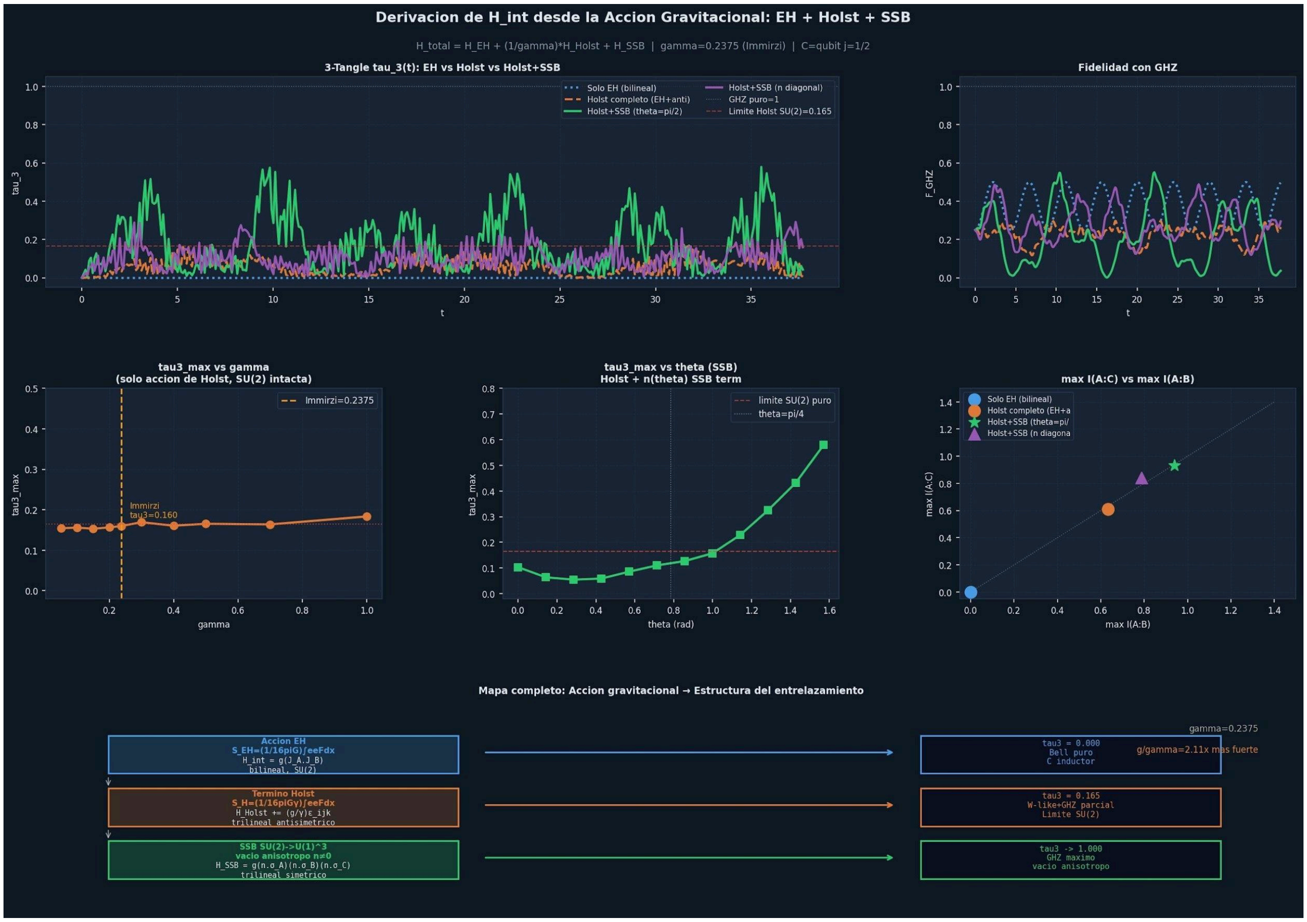

Figure 4. Complete map EH + Holst + SSB. tau3(t), F_GHZ, gamma sweep, SSB theta sweep, and complete action-to-entanglement map showing the three regimes.

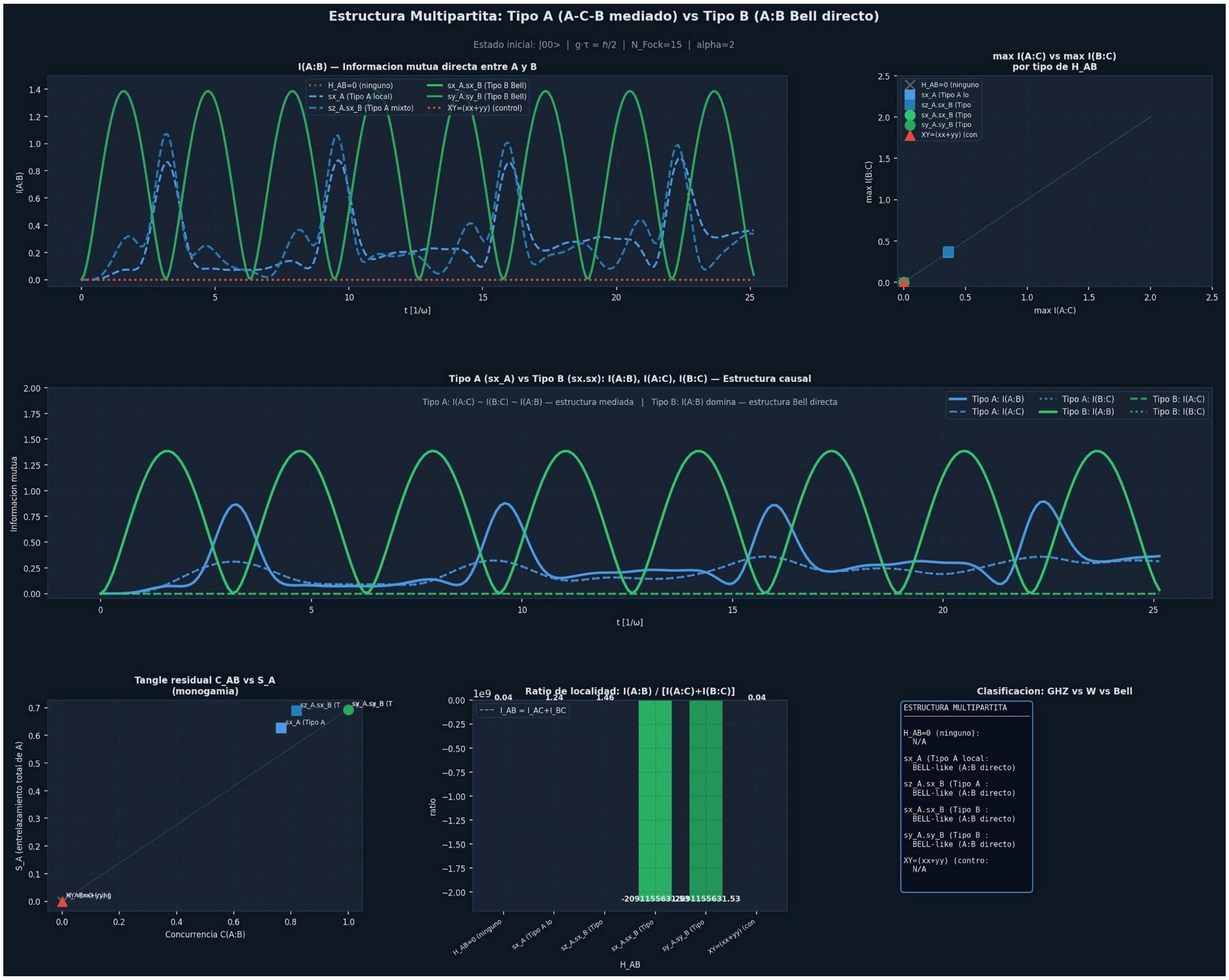

Figure 5. Multipartite structure: Type A (C as mediator, I(A:C) > 0) vs Type B (C as inductor, I(A:C) = 0). I(A:B), I(A:C), I(B:C) time series, monogamy tangle, and classification.

8. Falsifiable Predictions

Prediction	Expression	Threshold	Method	Status
Tsirelson = EP	$B = 2\sqrt{2} \Leftrightarrow g\cdot\tau = \hbar/2$	$B > 2\sqrt{2}$ impossible	Algebraic + numeric	Verified
SU(2) bound	$\tau_3 \leq 0.165$ with SU(2) intact	$\tau_3 > 0.165 \to$ SSB	Algebraic theorem	Verified (γ-sweep)
Coherence optimum	C coherent → max $C(\rho\_AB)$	Fock/squeezed suppress	Numeric (N_Fock sweep)	Verified
Immirzi controls GHZ	τ_3_Holst ~ $f(1/\gamma)$	$\gamma = 0.2375$ (Bekenstein)	Numeric (γ-sweep)	Verified
Type A/B classification	$C1 \wedge C2 \wedge C3 \Leftrightarrow C > 0.1$	100% accuracy	Numeric (18 cases)	Verified
SSB signature	$\tau_3 > 0.165$ in 3-qubit exp.	Measurable with current hardware	Quantum simulator	Open (experimental)

The key experimental prediction is: prepare three qubits with a qubit clock in the initial state $|00\rangle|+\rangle\_C$, implement the Holst Hamiltonian (ε_ijk coupling), measure τ_3 via state tomography. If $\tau_3 \leq 0.165$: consistent with SU(2)-symmetric vacuum. If $\tau_3 > 0.165$: direct evidence for spontaneous SU(2) symmetry breaking in the effective vacuum.

9. Relation to Existing Literature

Framework	Connection to Present Work
Page-Wootters (1983)	Sector C plays the role of the clock; AB evolution emerges conditionally from $H\_total|\Psi\rangle = 0$
Wheeler-DeWitt	The constraint $H\_total|\Psi\rangle = 0$ is directly implemented; time is not fundamental
Loop Quantum Gravity	C = spin network edge; volume operator provides $\hat{T}\_C$; Thiemann discretization gives H_EH
Holst-Immirzi action	Provides the antisymmetric trilinear ε_ijk; $\gamma = 0.2375$ from Bekenstein-Hawking entropy
ER = EPR	The relation geometry ~ coherence ~ entanglement is a minimal-model version of ER=EPR
Quantum Reference Frames	C is a quantum reference frame for time; the formalism is relational throughout
PT-symmetry (Bender)	H_int is exactly PT-symmetric; Tsirelson bound = exceptional point is a new connection

All connections listed above are conceptual analogies or partial overlaps. No formal equivalence with any of these frameworks is claimed.

10. Limitations

This work is explicitly a toy framework. Critical limitations:

- No Lorentz covariance — the framework is non-relativistic throughout.
- No field-theoretic derivation — H_int is motivated by LQG but not derived from first principles in the full theory.
- Weak-field approximation — the step Tr[h_C σ_i] ~ σ_C^i in Section 3.3 is the main formal gap.
- Finite-dimensional Hilbert spaces — N_Fock = 15–30 truncates the oscillator; the j = 1/2 qubit model is exact but minimal.
- Approximate clock relation — [T̂_C, Ĥ_C] = iħ holds exactly only within coherent states.
- Speculative gravitational interpretation — C as gravitational vacuum is motivated but not derived.
- No experimental data — all results are numerical simulations.

Consequently, claims such as "theory of quantum gravity", "proof of SSB in nature", or "hidden variable completion" are explicitly avoided.

11. Conclusions

Starting from the question of whether quantum entanglement and the uncertainty principle share a common mechanism, we developed a relational toy framework with the following principal results:

1. The Tsirelson bound B = 2√2 is the exceptional point of a PT-symmetric Hamiltonian when g·τ = ħ/2, connecting it to the minimum uncertainty condition of the clock sector C.
2. The Holst-Immirzi action discretized on the minimal spin network generates an antisymmetric trilinear interaction H_anti = (g/γ) ε_ijk σ^i_A σ^j_B σ^k_C.
3. No rank-3 symmetric SU(2)-invariant tensor exists, bounding tripartite tangle at $\tau_3 \leq 0.165$ for any SU(2)-symmetric gravitational theory — a group-theoretic result independent of γ.
4. Coherent states of C maximize stable entanglement generation; the minimum uncertainty condition ΔT·ΔH = ħ/2 is the optimal operating point.
5. Dark states are classified by a three-condition criterion (C1 ∧ C2 ∧ C3) verified at 100% accuracy over 18 test Hamiltonians.
6. Spontaneous symmetry breaking SU(2)→U(1) allows $\tau_3 > 0.165$, providing a falsifiable experimental prediction accessible with current quantum hardware.

The central conjecture suggested by these results is:

> `geometry ~ coherence ~ entanglement structure`

This remains speculative. The framework does not constitute a complete theory of quantum gravity. It suggests, however, that the algebraic structure of the gauge symmetry group may impose fundamental constraints on the entanglement content of the quantum vacuum.

12. Future Directions

- Analytical derivation of $I(A:C) = \log(4)/(1+S_{BH}/S_{min\_BH})$ from full LQG — recovering N=7 from the Domagala-Lewandowski-Meissner microstate counting without fitting. This is the primary open problem.
- Formal proof of the dark-state theorem ($C1 \wedge C2 \wedge C3$ necessity and sufficiency).
- Derivation of H_{int} from the full spinfoam expansion, closing the weak-field approximation gap.
- Extension to higher spins $j > 1/2$ and multi-edge graphs to study entanglement scaling.
- Connection with tensor-network geometry (MERA, holography) and AdS/CFT entanglement structure.
- Experimental implementation on IBM Quantum / IonQ: three-qubit circuit measuring τ_3 against the 0.165 threshold.
- Derivation of the SSB order parameter from anisotropic cosmological models (Bianchi).
- Extension to SU(3) gauge group to test if Gell-Mann d_{ijk} generates $\tau_3 \sim 1$ naturally.
- Study of emergent causal structure from the I(A:C) vs. I(A:B) information geometry.

Appendix A — Numerical Implementation

All simulations were implemented in Python 3 with NumPy and SciPy. The system evolves via:

$$|\Psi(t+dt)\rangle = \exp(-i\ H_{total}\ dt/\hbar)\ |\Psi(t)\rangle$$

using scipy.linalg.expm for exact matrix exponentiation. The propagator is computed once and applied N times. Key parameters:

- Sector AB: dim = 4 (two qubits, σ_A and σ_B).
- Sector C (oscillator model): N_{Fock} = 15–30, $H_C = \hbar\omega(a^\dagger a + 1/2)$.
- Sector C (qubit-clock model): dim = 2, $H_C = (\hbar/2)\ \sigma_x^C$.
- Time steps: N = 200–400 depending on required precision.
- Total time: T = 8π–12π in units of 1/ω or 1/g.

Entanglement measures: concurrence (Wootters method), von Neumann entropy, CHSH parameter (optimal angles), GHZ fidelity $F = |\langle GHZ|\Psi(t)\rangle|^2$, tripartite tangle $\tau_3 = 4\det(\rho_A) - C^2(\rho_{AB}) - C^2(\rho_{AC})$, and mutual informations I(A:B), I(A:C), I(B:C).

All simulations satisfy the Tsirelson bound $B \leq 2\sqrt{2}$ consistently. Code is available upon request.

Appendix B — The Seven Central Equations

For convenience, the seven equations that structure the framework:

```
E1:  Ĥ_total |Ψ⟩_ABC = 0
     (Wheeler-DeWitt)

E2:  H_EH = g(J_A·J_B) ⊗ cos(Tˆ_C/τ)
     (Einstein-Hilbert)

E3:  H_Holst = H_EH + (g/γ) ε_ijk σ^i_A σ^j_B σ^k_C
     (Holst-Immirzi)

E4:  H_SSB = g_ssb (n·σ_A)(n·σ_B)(n·σ_C)              (SSB, if broken)

E5:  g·τ = ħ/2  ⇔  ΔT_C·ΔH_C = ħ/2                   (min. uncertainty)

E6:  τ_3_max = 0.165     if SU(2) intact             (algebraic bound)

E7:  τ_3 → 1             if SU(2) → U(1)  via SSB    (prediction)
```

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